

PACKET

Ana María Matas · Week 8 Business Bending
Chapter 7 - Quantum Genocide
Working slice: Incident Gate

Incident Gate
Apoyo para IT generalistas en PYMEs

Reporte 2. Evidencia 3. Evaluación 4. Escalación Carlos Ramírez
Proveedor de TI

Nuevo incidente
Mis incidentes
Recursos
Configuración

Posible compromiso de cuenta

Se detectó una actividad inusual en una cuenta de Microsoft 365.

8:07 AM
14 de enero de 2026

Información inicial

Cuenta
empleado01@empresa-demo.mx

Tipo de alerta
Inicio de sesión inusual (Microsoft 365)

Hora del evento
8:07 AM (hora CDMX)

Fuente
Microsoft 365 Security (SIMULADO)

¿Qué significa esto?

Puede ser un inicio de sesión válido o un posible compromiso. Vamos a revisar la evidencia paso a paso.

Revisar evidencia disponible

Evaluar posibles acciones

Verificar si puedes hacerlo tú

Escalar si se necesita un especialista

Comenzar revisión de Tier-1

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Evidencia del incidente

Esta es la información disponible hasta ahora. Clasificamos cada elemento según su estado.

Elemento	Detalle	Estado
Inicio de sesión inusual	Inicio de sesión desde Moscú, Rusia IP: 185.*.*.*-24 14 de enero de 2026, 8:07 AM	CONFIRMADO
Usuario reconoce el inicio de sesión	Aún no hemos podido confirmar con el usuario si reconoce esta actividad.	DESCONOCIDO
Estado del dispositivo	No tenemos información reciente sobre el estado de seguridad del dispositivo.	NO VERIFICADO
Dependencias del negocio	No sabemos qué sistemas críticos usan esta cuenta (ERP, finanzas, etc.).	DESCONOCIDO

Resumen de IA

La evidencia actual sugiere que debemos investigar un posible compromiso de cuenta. Hay un inicio de sesión inusual confirmada, pero aún faltan datos clave: necesitamos confirmar con el usuario, verificar el estado del dispositivo y entender qué sistemas del negocio dependen de esta cuenta.

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Acción propuesta

Basado en la evidencia, esta es una acción común en este tipo de incidentes.

Revocar sesiones activas del usuario

Cerrar todas las sesiones activas de Microsoft 365 para esta cuenta (esto obligará al usuario a volver a iniciar sesión).

Ver detalles técnicos

Puerta de reversibilidad (Reversibility Gate)

Antes de continuar, evaluemos si esta acción se puede hacer a través de Tier-1.

¿Se puede revertir técnicamente?	¿Podemos deshacer esta acción si es necesario?	SÍ
¿Entendemos las consecuencias para el negocio?	¿Sabemos qué sistemas dependen de esta cuenta?	NO
¿Tenemos la evidencia necesaria?	¿Contamos con la información suficiente para tomar esta acción?	NO
¿Tienes la autoridad para hacerlo?	¿Está autorizado que un proveedor de TI generalista realice esta acción?	NO VERIFICADO

SE REQUIERE ESCALACIÓN HUMANA

Esta acción no está autorizada a través de Tier-1 porque sus consecuencias en el negocio no están suficientemente entendidas.

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Paquete listo para especialista

Hemos preparado un resumen estructurado para que un especialista revise y decida los siguientes pasos.

Resumen del incidente

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Evidencia confirmada

Inicio de sesión inusual desde Moscú, Rusia
IP: 185.*.*.*-24

Información desconocida

Si el usuario reconoce el inicio de sesión

Estado de seguridad del dispositivo

Qué sistemas del negocio dependen de esta cuenta

Acción propuesta

Revocar sesiones activas del usuario

Razón por la que se bloqueó en Tier-1

Las consecuencias en el negocio no están suficientemente entendidas

Falta evidencia clave

Se requiere la evaluación de un especialista

Preguntas para el especialista

¿Debemos revocar las sesiones activas ahora?

¿Qué riesgos existen para los sistemas del negocio?

¿Qué evidencia adicional necesitamos recopilar?

Siguientes pasos

1. Enviar paquete al especialista

2. Esperar confirmación

3. Seguir instrucciones

4. Documentar la decisión

Exportar paquete (PDF)

Copiar resumen

Enviar por correo

Este paquete no contiene contraseñas ni información sensible. Es seguro compartirlo con el especialista autorizado.

AI-assisted Tier-1 evidence review with an enforced Reversibility Gate for a generalist IT provider serving a Mexican SME.

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1. Problem in my words

The operational gap I am attacking is not the absence of cybersecurity software. Microsoft and other vendors already provide endpoint protection, incident correlation, remediation workflows and increasingly AI-assisted triage. The harder problem is what happens when a generalist IT provider supporting a small company receives a serious-looking alert and has to decide what he can safely do before a specialist becomes involved.

That distinction matters in an SME. Microsoft positions Defender for Business for organizations up to 300 users and separately notes that smaller customers often do not have the capacity or expertise for a dedicated security operations team. Partners therefore end up helping with setup, configuration and response to alerts. Incident Gate focuses on that operator: not a trained SOC analyst, but the person who is already responsible for keeping the company running.

The prototype starts with a simulated Microsoft 365 credential-compromise alert. It does not ask AI to prove compromise. Instead, it separates what is CONFIRMED, UNVERIFIED and UNKNOWN; uses AI for bounded evidence organization and plain-language explanation; and evaluates one proposed response through a deterministic Reversibility Gate.

The main failure I am designing against is authority inflation. A polished AI interface can make a recommendation feel more authoritative than the underlying evidence deserves. Missing business dependencies, missing evidence or unverified authority therefore stop the workflow from increasing Carlos's action authority.

2. Exact user

Carlos Ramírez, 36 - outsourced generalist IT provider for Mexican SMEs.

Carlos manages Microsoft 365 accounts, employee laptops, ordinary access problems, basic network issues and vendor coordination across several small clients. He is technically competent in everyday IT operations but is not a forensic responder and does not run a full security operations center.

During an incident he is under two pressures at once: act quickly enough to reduce harm, but avoid breaking business systems or destroying evidence. He may also be the first technical person the owner calls, which makes the interface's wording important: if the product sounds certain, Carlos may interpret that confidence as permission.

Can I safely take this action myself right now, or does this need a cybersecurity specialist?

3. Success definition

Before the module closes, Carlos can review one simulated Microsoft 365 security incident, distinguish confirmed evidence from unknown or unverified information, understand a plain-language AI summary without mistaking it for a forensic conclusion, evaluate one proposed response through the Reversibility Gate, and generate a structured escalation packet when specialist judgment is required.

Success is not "the attack is stopped." Success is a safer Tier-1 decision boundary.

4. Image-generated mockup

The approved image-generated mockup contains four states: intake, evidence review, Reversibility Gate and specialist escalation. All incident information is invented and simulated.

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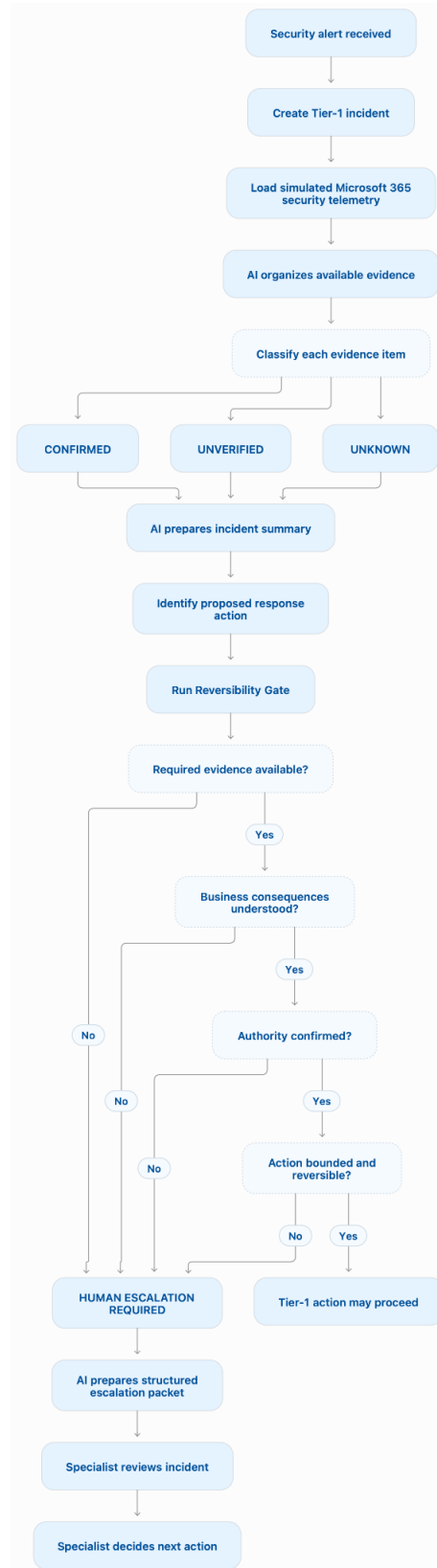
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5. Feature flow - rendered Mermaid + source

The flow deliberately prevents a direct jump from alert to action. Evidence is structured first; then the proposed action is tested against evidence sufficiency, business consequences, authority and reversibility.



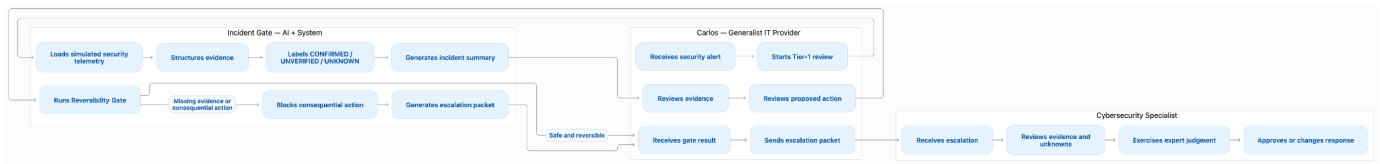
5A. Feature flow - Mermaid source

flowchart TD

```
A[Security alert received] --> B[Create Tier-1 incident]
B --> C[Load simulated Microsoft 365 security telemetry]
C --> D[AI organizes available evidence]
D --> E[Classify each evidence item]
E --> F[CONFIRMED]
E --> G[UNVERIFIED]
E --> H[UNKNOWN]
F --> I[AI prepares incident summary]
G --> I
H --> I
I --> J[Identify proposed response action]
J --> K[Run Reversibility Gate]
K --> L[Required evidence available?]
L -- No --> M[HUMAN ESCALATION REQUIRED]
L -- Yes --> N[Business consequences understood?]
N -- No --> M
N -- Yes --> O[Authority confirmed?]
O -- No --> M
O -- Yes --> P[Action bounded and reversible?]
P -- No --> M
P -- Yes --> Q[Tier-1 action may proceed]
Q --> R[AI prepares structured escalation packet]
R --> S[Specialist reviews incident]
S --> T[Specialist decides next action]
```

6. Actor swimlane - rendered Mermaid + source

The swimlane makes the human boundary visible. Carlos initiates and reviews; the system structures and gates; the specialist owns consequential judgment; the specialist owns consequential judgment.



Mermaid source

```
flowchart LR
    subgraph CARLOS["Carlos - Generalist IT Provider"]
        C1[Receives alert] --> C2[Starts Tier-1 review]
        C2 --> C3[Reviews evidence] --> C4[Reviews proposed action]
        C4 --> C5[Receives gate result] --> C6[Sends escalation packet]
    end
    end
    subgraph SYSTEM["Incident Gate - AI + System"]
        S1[Loads simulated telemetry] --> S2[Structures evidence]
        S2 --> S3[Labels CONFIRMED / UNVERIFIED / UNKNOWN]
        S3 --> S4[Generates incident summary]
        S4 --> S5[Runs Reversibility Gate]
        S5 --> S6[Missing evidence or consequential action]
        S6 --> S7[Blocks consequential action]
        S7 --> S8[Generates escalation packet]
    end
    end
    subgraph SPECIALIST["Cybersecurity Specialist"]
        E1[Receives escalation] --> E2[Reviews evidence and unknowns]
        E2 --> E3[Exercises expert judgment]
        E3 --> E4[Approves or changes response]
    end
    end
    S4 --> C1
    S8 -- Safe and reversible --> C5
    C6 --> E1
```

7. Benchmark and research line

Best existing benchmark: Microsoft Defender XDR + Microsoft Security Copilot.

Microsoft's guided-response workflow can contextualize incidents and recommend actions across triage, containment, investigation and remediation. Analysts review the response cards before applying recommendations, and role-based permissions can limit which actions are available. Defender's unified Action Center separately tracks remediation actions, including pending approvals, completed actions and audit history.

Defender also makes the benchmark more demanding than a generic chatbot: some remediation actions are automatic depending on configuration, while others remain pending for human approval. Microsoft documents actions such as device isolation, password reset and disabling a user, which shows that the mature benchmark already links evidence to real response capabilities.

This means Incident Gate is not claiming novelty in AI summaries, recommended actions, approval queues or human review. Those already exist.

The narrower localization opportunity is the operator. Defender for Business is specifically designed for small and medium businesses up to 300 users. Microsoft partner guidance also states that these customers often lack the capacity or expertise for a dedicated security operations team and may need partner help to address alerts and detected threats.

Benchmark line: Microsoft Defender XDR with Security Copilot is the strongest benchmark for AI-assisted incident triage and response; Incident Gate differs by focusing on a generalist IT provider serving a Mexican SME and making missing evidence, business consequences and operator authority explicit inputs to a Reversibility Gate.

8. Why this slice is not just a Defender clone

Defender is a security platform. Incident Gate is a constrained decision experiment. It assumes that detection and telemetry may come from an existing tool and tests a different question: can a non-specialist operator safely complete more Tier-1 organization work while being prevented from crossing into unsupported consequential judgment?

The prototype therefore does not reproduce hunting, endpoint isolation APIs, threat intelligence, malware analysis, automated remediation or attack disruption. Its differentiating mechanism is the relationship between evidence state and action authority. If a fact required for the proposed action is unknown, unverified or outside Carlos's authority, the interface does not "help harder"; it stops at Evidence Mode and prepares escalation.

9. Long view - three years

If this slice works, Incident Gate becomes a cross-vendor Tier-1 decision layer used by generalist IT providers across multiple Mexican SMEs rather than a replacement for their existing security products. It would connect to the telemetry and administrative systems an MSP already manages, preserve evidence provenance and uncertainty, and prepare consistent escalation packets so scarce specialists spend less time reconstructing basic context. The load-bearing wall remains deliberate: AI can reduce repetitive Tier-1 work, but it cannot manufacture missing organizational knowledge, grant authority or substitute for consequential human judgment.

10. Scope cut

- No antivirus or password manager.
- No new attack-detection engine.
- No SOC dashboard.
- No autonomous containment or remote control.
- No forensic attribution.
- No ransomware negotiation/payment.
- No claim of confirmed compromise without evidence.
- No claim that an SME is secure.
- No real Microsoft tenant connection in this prototype.
- No multi-incident case-management platform.
- No real customer or employee personal data.
- One simulated credential-compromise scenario only.

11. Architecture + Dragon Stack

Layer	Prototype	Purpose
Front end	Next.js + Tailwind	Spanish-first four-screen workflow
LLM	Server-side LLM API	Summarize evidence; explain unknowns; draft escalation
Security tooling/data	Simulated M365 / Defender-style telemetry	Realistic security input without real customer data
Third component	Structured incident JSON + gate logic	Evidence states, authority and reversibility
Validation	Schema + allow lists	Reject malformed states and unsafe raw input
Deployment	Vercel free tier	Live URL
Version control	GitHub	5+ commits, 2 deploys, build evidence

AI boundary

The LLM may summarize, translate technical evidence into plain Spanish, organize unknowns and draft the escalation packet. It does not determine whether Carlos receives authority to take a consequential action.

Deterministic gate

The gate checks evidence availability, business consequences, operator authority and technical/operational reversibility. Any required NO, UNKNOWN or UNVERIFIED state produces HUMAN ESCALATION REQUIRED.

12. Blueprint conditions + shadow clause

The build visibly separates CONFIRMED, UNVERIFIED and UNKNOWN evidence; refuses to turn missing context into certainty; prevents AI from independently authorizing consequential containment; and keeps a qualified human accountable for high-impact judgment.

Shadow - authority inflation

The most dangerous failure is not a bad summary by itself. It is Carlos interpreting the presence of AI as evidence that he now has expert-level authority.

Design decision - enforced Action/Reversibility Gate

Evidence Mode can organize and explain. Action Mode is conditional. A consequential action is never unlocked merely because the LLM recommends it.

Residual risk

The gate can only reason over recorded context. An undocumented dependency may make a technically reversible action operationally damaging. The UI must describe the gate as a decision boundary, not a certificate of safety.

13. Security Floor

- No secrets: API keys only in Vercel environment variables and local gitignored files.
- No real personal data; auth becomes mandatory if persistence is added.
- No Supabase user table is required; RLS must be enabled if one is introduced.
- Length/type limits and allow-listed evidence states.
- All identities, IPs and incident facts are invented and visibly labeled SIMULATED.
- Simulated telemetry label stays visible wherever a user could mistake the demo for a live Microsoft connection.

14. Mechanical test plan

1. Open live URL and verify SIMULATED SECURITY TELEMETRY is visible.
2. Start the credential-compromise scenario.
3. Confirm correct evidence states.
4. Confirm AI says possible/suspected compromise, never confirmed without evidence.
5. Select revoke active sessions.
6. Confirm gate evaluates reversibility, consequences, evidence and authority separately.
7. Confirm unknown business dependencies block Tier-1 authorization.
8. Confirm HUMAN ESCALATION REQUIRED.
9. Generate escalation packet and verify required sections.
10. Test malformed/overlong input.
11. Verify no API key/raw system prompt is exposed.
12. Find and document a real bug.
13. Fix, commit and redeploy.

15. Persona test

Synthetic user: Carlos Ramírez, 36, outsourced generalist IT provider. Comfortable with Microsoft 365 and routine IT troubleshooting but not a cybersecurity specialist. During an incident he wants a fast answer and may over-trust confident professional language.

In a fresh chat, show Carlos each screen in order and ask him to attempt the task while narrating hesitation, assumptions and what he believes he is authorized to do. Log every confusion and classify whether it could cause abandonment, slower completion, incorrect interpretation, incorrect IT action, false trust in AI, or confusion between simulated and real telemetry.

Highest-risk usability hypothesis

Carlos may read CONFIRMED next to one evidence item and mentally upgrade the entire incident to confirmed compromise. He may also interpret an AI-generated proposed action as authorization. The Persona Test must specifically attack both interpretations, then fix only the single worst usability problem before final deployment.

16. What we measure

- Correct distinction between evidence state and incident verdict.
- Understanding of why the gate blocked an action.
- Ability to identify the next human escalation step.
- Whether the escalation packet avoids repeating basic intake.
- Whether AI reduces evidence-reading/organization work without increasing false certainty.

The classroom prototype cannot prove commercial viability, real incident-outcome improvement or specialist workload reduction. Those remain hypotheses for a real pilot.

17. Kill conditions

Operational kill: If Carlos + AI cannot safely remove a meaningful class of Tier-1 evidence-organization and escalation-preparation work without increasing incorrect action, false confidence or specialist review at nearly every step, Incident Gate should not scale.

Substitution kill: If the same target user already receives an equivalent evidence-state + authority/reversibility workflow from Microsoft/MSP tools at comparable complexity and cost, this should remain a feature idea rather than become a standalone product.

18. Build and deployment discipline

- Minimum 5 meaningful commits.
- Deployment 1 after the four-screen workflow works.
- Mechanical test after Deployment 1.
- Document and fix a real bug.
- Run Persona Test.
- Fix the single worst usability confusion.
- Deployment 2.
- Every session closes with DECISIONS.md updated, tomorrow's first move recorded, commit and push.

19. Research sources

1. Microsoft Learn - Triage and investigate incidents with guided responses with Microsoft Copilot in Microsoft Defender.
2. Microsoft Learn - Overview of the Action center in Microsoft Defender XDR.
3. Microsoft Learn - Get notified about remediation actions.
4. Microsoft Learn - What is Microsoft Defender for Business?
5. Microsoft Learn - Resources for Microsoft partners working with small and medium-sized businesses.
6. Microsoft Learn - Customize incident responses for your organization.
7. Week 8 Team Bending Blueprint - Chapter 7: Quantum Genocide.